

FCIT23

CPCS-204 Final Exam

2024 Dec 19th

شكراً لكل اللي جمعوا الاختبار لا تنسونهم من دعائكم (:

Question 1:

Write a recursive java method named reverse() that takes a String ,

The method then swaps the first character with last character , and then second character with second last character and so on.....

Examples :

ABCDE --> EDCBA

Question 2:

Write a method in java to calculate the sum of the last two nodes in a singly linked list.

The linked list must have at least two nodes; otherwise, return -1

Examples:

1. For the linked list: 1 -> 2 -> 3 -> 4, the method should return 7 (3 + 4)
2. For the linked list: 1 , the method should return -1 (less than two nodes)
3. For an empty list (null) , the method should return -1

Code :

```
public int sumLastNode( llnode head){
// Check if the list has less than two nodes
if (.....){
return-1:
}
llnode pre == null;
llnode current = head;

// Traverse the list
while ( .....){

..... = current;
.....

}
// Return the sum of the values of the last two nodes
return .....;
}
```

Question 3 :

Write a method called `isIdentical` that returns `True` if the nodes in the tree are equal

Code:

```
boolean isIdentical(TreeNode p1, TreeNode p2) {  
    // Base case: both nodes are null, trees are identical  
    if (.....) {  
        return true;  
    }  
    // If one node is null, or their data is not equal, trees  
    //are not identical  
    if (.....) {  
        return false;  
    }  
    // Recursively check left and right subtrees  
    return .....  
}
```

Question 4 :

Given the recursive method below , what would this method return when its called num(1234)?

Draw the recursion trace showing all the recursive calls and results .

*the trace can also be represented as a binary tree.

Code :

```
public static int Split(num){  
    if (num<10)  
        return num;  
  
    int leftpart = num/10;  
    int rightpart = num%10;  
    return split(leftpart) + num +  split(rightpart)  
}
```

Trace:

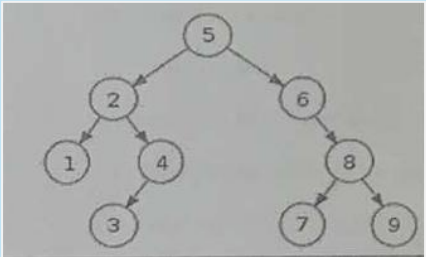
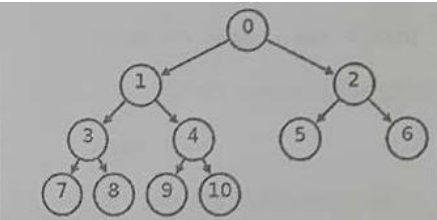
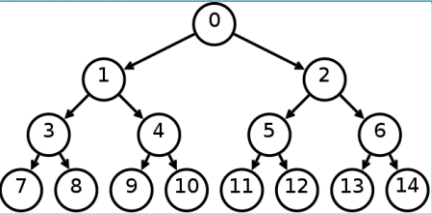
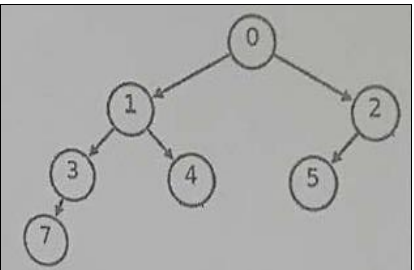
Result:.....

Question 5 :

Algorithm A runs in $O(n^3)$ time. For an input of size 4 , the running time 192 milliseconds. How long will Algorithm (A) take to run on an input of size 8?

Question No.6 :

Distinguish types of tree , choose all that applies , write(yes/no):

Tree Structure	BST	Full binary tree	Complete binary tree
			
			
			
			

Question No.7:

Do the merge sort for the following array , show all the steps :

38	27	43	3	9	82	10	42
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Question No.8:

Convert the following infix expression to postfix expression using stacks :

$$3 * (2 + (9-6) / 3) + (4*2)$$

Postfix expression :

Question No.9 :

Let Q be a queue initially having the values 20,40,60,80,and 100 , with 20 in the front. Also assume that S be a stack initially having the values 11 , and 12 , with 12 at the top. What the following fragment code would do?

```
for (int i = 0 ; i < 2 ; i++){  
    System.out.print(q.dequeue) ;  
    System.out.print (s.pop) ;  
    Q.enqueue(s.top+i) ;  
    S.push(q.front) ;  
}
```

Output :

Countent of queue :

Content of stack :

Question No.10 :

Selection sort :

i	key		2	4	10	3	1	7	5

Question No.11 :

Determine the big-O running time of the following code fragment, you must explain your answer

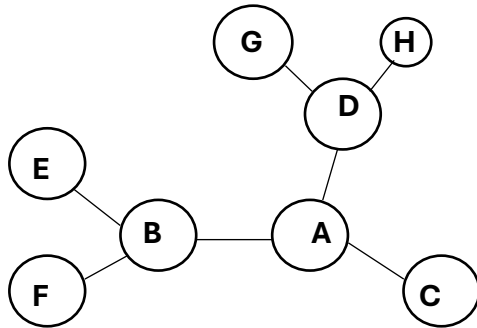
```
int exam(int n){
    int i = 0, j = 0;
    while(i < n){
        j = 0;
        while (j < n) {
            System.out.println("wow");
            j++;
        }
        i++;
    }
    return n;
}
```

Solution :

Explanation :

Question No.11 :

Graph: do a DFS on the following graph. Staring from the node “A” and ensuring you follow the alphabetical order when selecting the next node to visit



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Question 12: Hash Table

Consider the following list 12, 19, 25, 32, 37. And the following two functions.

$$H1(X) = (3x + 7) \bmod 13$$

$$H2(X) = 7 - (x \bmod 7)$$

i	Double hashing	Separate chaining
0		
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		

Question 13 (MCQ): مب كلها زي الاختيار والخيارات اللي بالاختبار بعضها مقارب

1- What does the Selection Sort algorithm do?

- A. Finds the largest value and swaps it with the first position.
- B. Repeatedly compares adjacent elements and swaps them if needed.
- C. Divides the list into smaller parts and sorts them recursively.
- D. Swap randomly

2- Which sorting algorithm requires intermediate space for its operation?

- A. Merge Sort
- B. Quick Sort
- C. Selection Sort
- D. Insertion Sort

3- What would be the best pivot in 9, 12, 18, 7, 9?

- A. 19
- B. 7
- C. 12
- D. It doesn't matter

4- In Sorted array which operation is fastest?

- A. Quick
- B. Selection
- C. Insertion //-----:
- D. Bubble

5- What is the best data structure to implement BFS in a graph?

- A. Linked list
- B. Stack
- C. Queue
- D. None

6- Given the following operation on the Stack: Push (1), Push (2), Pop (), Push (3), Push (4), Pop (), what is the top element?

- A. 1
- B. 2
- C. 3
- D. 4

7- Which of the following scenarios is an example of stack?

- A. A line of people waiting for bus
- B. The undo function in a word processor
- C. Retrieving a book from the bottom of a stack
- D. Sorting letter in alphabetical order

8- Which operation is used to remove an element from the stack?

- A. Push
- B. Pop
- C. Peek
- D. Clear

9- What operation behaves like opening a new page after going back multiple times, effectively removing everything? ما كان كذا السؤال كان مفخخ أكثر

- A. Clear
- B. Push
- C. Pop
- D. Peek

10- What is the purpose of the adjacency matrix in a graph?

- A. To track the neighbors of each node
- B. To store the neighbors along with their weights
- C. To store the degree of each node
- D. To represent the structure of the graph as a tree

11- What kind of data structure is used to implement Merge sort?

- A. Stack
- B. Queue
- C. Array
- D. Linked List

12- Which of the following conditions causes the worst-case time complexity of Quik Sort?

- A. The pivot divides the array into two equal parts
- B. The pivot is always the largest of smallest elements
- C. The array has duplicate elements
- D. None of the above

Answers

1-A, 2- A, 3-C, 4-C, 5-C, 6-C, 7-B, 8-B, 9-A, 10-B, 11-C, 12-B